Bounce-apart calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum and energy to check an elastic collision.

SUCCESS CRITERIA - I can...

1. Write a signed momentum balance.
2. Check kinetic energy separately.
3. Explain why a collision cannot be elastic if its total kinetic energy changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two equal **3 kg** carts collide elastically along one line. A starts at **+5 m/s**; B starts at rest. **Find both final velocities** in this ideal case.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{total,\,i} = p_{total,\,f}$; $K_{total,\,i} = K_{total,\,f}$.
Isolated elastic collision; $K = \frac{1}{2}mv^2$. Equal masses exchange velocities in the ideal one-dimensional case.

Worked reasoning: A **1 kg** cart moving at **+3 m/s** hits a stationary **2 kg** cart. The first rebounds at **-1 m/s**. Total momentum was **+3 kg·m/s**, so the second must have momentum **+4 kg·m/s** and velocity **+2 m/s**.

Your turn: Draw momentum arrows before and after; verify the total.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Equal-mass carts collide head-on elastically. A starts right at **3 m/s**; B rests. **Sketch the after state**.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A **1 kg** cart at **+4 m/s** hits a **3 kg** cart at rest. After an ideal elastic collision A is at **-2 m/s**. **Use momentum** to **find B velocity**, then verify kinetic energy. Organize the known quantities on the before-and-after diagram.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

Bounce-apart calculations

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Continue down each column.

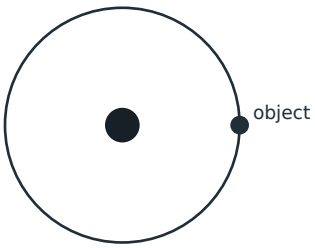
05 Explain your reasoning

A pair’s total momentum matches before and after, but kinetic energy falls. Is the collision elastic? Explain **0**.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A circular orbital radius becomes **16** times as large around the same source. **Compare orbital speed** and period. **Label velocity** and inward acceleration on the circular-path diagram.



Marzano check - circle: 1 2 3

07 Exit check

A **1 kg** cart at **+3 m/s** and **1 kg** cart at **-1 m/s** stick. Compute final velocity and **explain why this** is not the elastic result. Organize the known quantities on the before-and-after diagram.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** cart at **+5 m/s** hits a **3 kg** cart at **-1 m/s**. The first leaves at **-2 m/s**. **Find the second's final velocity**, and the impulse on each cart. Organize the known quantities on the before-and-after diagram.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3